Topology-Aware Differentiable Triangle-Soup Reconstruction via Persistent Homology

Anonymous 3DV submission

Paper ID *****

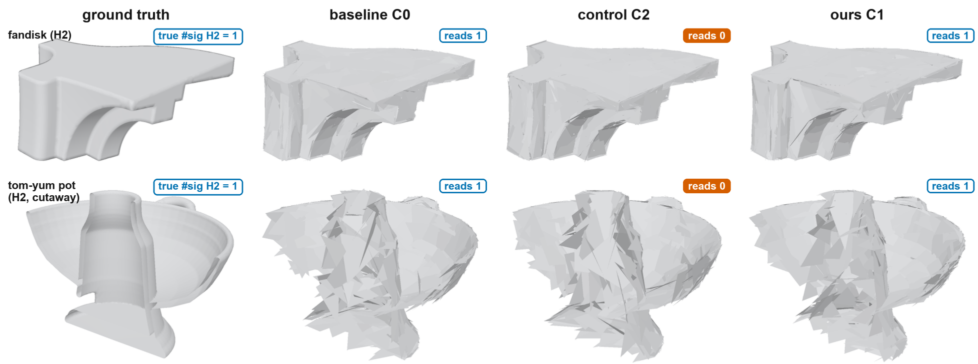

Figure 1. Geometry cannot carry the verdict — measurement can. Final reconstructions (seed 0), identical budgets: ground truth; baseline (C0); the norm-matched non-topological repulsion control (C2); the persistence loss (C1, ours) — equal-or-better Chamfer on every row (§5.2). A topological error need not be visible, so each cell is badged with its measured significant-feature count (blue outline = correct, filled orange = wrong; identical across seeds): the control erases both voids — and collapses a loop on bob (suppl. Fig. S8) — while the loss reads correct everywhere, cutting tail error up to 10.4× (group means 1.5×/4.9×; Tables 1–2).

Abstract

Differentiable triangle-soup reconstruction inherits a blind spot from its objective: photometric and geometric losses cannot see topology, so a reconstruction with a collapsed loop or a punctured enclosed void can score exactly as well as a correct one (on Chamfer-equal probes the diagrams differ 35–40× in bottleneck distance). The standard implicit remedy — steer where the resampler spends its budget — does not repair this: in a controlled allocation study, a topology-informed prior is largely matched by an equally wide random one, and no prior shape repairs loops. We therefore move topology into the objective: a differentiable persistence term compares the evolving surface's diagram, measured on live surface samples, to a fixed target; gradients flow through a pair-frozen backward re-expressing matched birth/death simplices as closed-form circumradii (exact for Gabriel simplices, verified per pairing), plus a recruitment term restoring the gradient optimal matching provably lacks when a target feature is missing; one ratio knob calibrates the loss against the photometric gradient, no curriculum needed. Every claim passes a channel-controlled verdict: the loss must beat a norm-matched non-topological control through the identical gradient channel, at Chamfer parity. Under that rule the loss is topology-specific for enclosed voids (4.0–7.9× lower error) and — the class every allocation prior failed — for loops (2.3×, zero phantom handles, while the control collapses one); loss and prior compose; component counts (H0) are an honest null, matched by the generic control. The verdicts replicate without per-shape tuning on eight external genus-known meshes in two observable-class groups (pre-registered group means: loops 1.52×, voids 4.87×; one a raw artist mesh, ingested and certified; the one non-pass is a no-headroom null, its already-correct baseline left unharmed), degrading gracefully under sensor noise. Scope: all evidence is synthetic and single-machine — closed surfaces plus a first open-surface probe — with the target diagram known in advance, the regime where correction can be measured exactly; real scans and blind capture are future work. Prescription: correct topology in the loss, allocate wide, combine.

1. Introduction

Differentiable rasterization reconstructs a triangle soup from images by following photometric gradients [47], periodically resampling under a fixed budget. Photometric losses are topologically blind: reconstructions differing in genus, connectivity, or enclosed voids can be indistinguishable to Chamfer and Hausdorff while persistence diagrams separate them 35–40× in bottleneck distance on Chamfer-equal probes (§2; suppl. Table S1). Fig. 1 previews the consequence: identical budgets, equal-or-better Chamfer, topology standing or falling with the objective.

A decade of connectivity-free reconstruction — point splats, Gaussians, and lately a return to triangle soups for their native graphics-pipeline fit [6, 21, 47] — has treated topology errors as symptoms of geometric under-provision: represent the surface better, or allocate the budget more wisely, and topology should follow. We tested the allocation half first (suppl. §A). It fails: a topological prior helps enclosed voids, but any equally-wide non-topological prior reproduces most of the gain, and no prior shape repairs loops — concentrating manufactures phantom handles, spreading never beats a random control.

The null result relocates the root cause: not where triangles are spent, but what the objective measures — a photometric loss is a local, pixel-space signal; a loop or an enclosed void is a global, categorical property of the surface. This paper therefore moves topology into the objective: a differentiable persistence term whose gradients act directly on features' birth/death values rather than on where triangles respawn.

Three research hypotheses structure the study, each tested by a pre-designed condition:

1. RH1 (diagnosis): topology errors originate in the training objective, not in primitive allocation — tested by the allocation study above.
2. RH2 (specificity): a persistence term improves topology in ways no norm-matched non-topological regularizer through the identical channel can reproduce — the control C2 (§3.5).
3. RH3 (complementarity): objective and allocation are complementary channels, not substitutes — the composition C5.

Concretely, we contribute a matched persistence loss with a pair-frozen backward (§3): exact persistence forward (GUDHI alpha complex), backward through closed-form circumradii — Gabriel-exact for 100% of significant pairs in practice; a recruitment term for missing features: optimal matching yields zero gradient when a target feature has no live counterpart; recruitment attaches the nearest live bar, and removing it collapses the loop-class win to baseline (§5); one calibrated knob: λ set by a gradient-norm ratio ρ, flat over a 10× range, no curriculum needed — an ablation, not an assumption; and channel-controlled evaluation: every claim must beat a norm-matched non-topological control through the identical channel (§3.5).

The loss proves topology-specific on the class no prior ever won — loops, zero phantom handles — confirming RH2; the channels compose (RH3); the H0 class is matched by the generic control, sharpening RH1: topological guidance earns its keep where geometry and topology decouple.

2. Related work

A decade of gradient-based reconstruction: four positions on connectivity. OpenDR [33], neural mesh rendering [27], SoftRas [32], DIB-R [8] and nvdiffrast [29] established gradient flow from pixels to geometry; reconstruction work since splits on what that geometry should be. (i) Implicit fields — learned occupancy and signed distance [35, 40], volume-rendered from NeRF [36] through NeuS [50] to Neuralangelo [31] — are watertight and topologically consistent by construction (the strongest counter-position), at field-evaluation and extraction cost. (ii) Fixed-connectivity templates deformed per scene [19, 49] keep manifoldness but freeze topology at the template's genus. (iii) Differentiable iso-surfacing re-extracts the mesh from a volumetric scaffold at every step [37, 44, 45] (Shape-as-Points bridges point soups to watertight implicits [41]) — topological freedom at grid cost. (iv) Connectivity-free primitive soups — point splatting [51], patch atlases [17], Gaussians [28] and their surface-accurate variants [18, 24], and a 2025–26 return to native triangles (splatted [21], soft-connected [6], rasterized by DiffSoup [47]) — are the fastest and the easiest to budget, but even their surface-minded members regularize geometry (alignment, flatness, connectivity forces), never a measured topology. MeshSplatting, conversely, renders already-connected opaque meshes differentiably [20]: connectivity as input; ours a free soup with measured topology. (i) and (iii) solve topology representationally; (iv) had not addressed it. Cut by optimization philosophy rather than representation, the decade holds geometry-oriented objectives (every method above), sampling-oriented steering of where a fixed objective spends primitives (below), and an empty third family — topology-oriented objectives that measure the property and differentiate through it (suppl. Table S6). Prior work occupies every pairwise intersection of the section's axes — explicit triangles, differentiable optimization, topology in the objective; this paper takes the three-way one, at the soup's budget and speed. We build on DiffSoup, whose in-loop prune/respawn resampler is the very mechanism our allocation study (suppl. §A) steered; here the resampler stays untouched and only the objective changes. Vertex-domain preconditioning [38] is complementary: it changes how vertices move, we change why.

The allocation channel: why priors alone fail. Three findings of our allocation-channel study set this paper's problem. (1) Geometric metrics are topology-blind: on bisection-matched probes with Chamfer equal by construction (0.916/0.628/0.977% per pair), the discriminating-dimension bottleneck separates correct from broken topology by 35–40× — thin handle ≈0 vs .0302, bridged merge .0014 vs .0574, punctured void .0040 vs .1385 — and Hausdorff$_{95}$ even prefers the wrong candidate in two of three cases (suppl. Table S1). (2) For voids, a spread topological prior cuts the bottleneck 33–53% below baseline — but a width-matched random control recovers most of that gain, leaving a small, shape-dependent residual: the prior's value is carried largely by its width. (3) No prior shape repairs loops: the best torus arm is the non-topological wide control (.0267 vs the spread prior's .0316), and the concentrated field manufactures phantom handles (4.4 significant loops vs the true 2); C5 below reuses the study's strongest field (B4; suppl. §A).

Persistence in learning objectives. Persistent homology [13] with its stability guarantees [10] has been used as a trainable signal via differentiable sorting of filtration values [5], for topology-preserving image segmentation [9, 23], point-cloud and shape optimization and 3D volume prediction [4, 42, 48], and diagram-metric optimization with convergence analysis [7]; barcode-valued maps have since been given a general differential framework [30], and the per-step cost of persistence-guided descent attacked directly [39] — yet never as the objective of an explicit triangle-based reconstruction. Nearest our setting, concurrent work brings persistence to reconstruction at other, complementary entry points: an image-space barcode term for Gaussian splatting [46], homology-guided camera placement [15] or a genus-matched template [14] in mesh inverse rendering, and a connected-component constraint on implicits fitted to point clouds [25]. Partition our setting's three requirements — (i) a topological quantity of the evolving surface measured differentiably in the objective, (ii) image-based training through a renderer, (iii) a fixed primitive budget — and each neighbour satisfies at most one (suppl. §D.7): the image-space barcode term never measures the surface (i); the point-cloud pipeline forgoes renderer and budget (ii, iii); the template line assumes the genus we measure (i). A head-to-head benchmark is therefore ill-posed (inputs, outputs, and cost regimes differ); our evaluation instead pits the loss against norm-matched controls through its own channel (§4). The nearest soup-side neighbour regularizes coherence with soft connectivity forces [6] — local continuity pressure, no measured topology; we replace that heuristic with the measurement. Our backward pass is in the persistence family — unlike work that back-propagates through filtration-value sorting or barcode coordinates in general position, we re-express each paired simplex's birth/death as a closed-form circumradius — with three pipeline-specific ingredients. (i) Alpha-complex pairing over $\alpha\times$area surface samples of a live triangle soup — not image grids or raw clouds — couples the diagram to the geometry actually being trained. (ii) A Gabriel-exactness gate: the circumradius re-expression equals the filtration value only for Gabriel simplices, so we verify it per pairing (relative 10^{-6}) — certifying the frozen gradient rather than assuming it. (iii) Recruitment for missing features: optimal partial matching has zero gradient exactly when a target feature is entirely absent, so we trade metric fidelity for a restored descent direction (§3), quantifying both its pathology and its necessity (§5).

Topology-aware geometry processing. Topology-aware reconstruction and repair predate differentiable pipelines [2, 26, 43], and budgeted remeshing has a long lineage [1, 3, 11, 16, 22]; our contribution is orthogonal — the budget mechanics stay DiffSoup's, only the training signal becomes topology-aware.

3. Method: a matched persistence loss for triangle soups

3.1. Objective

DiffSoup trains with a photometric objective L_{photo} (opacity auxiliary $+\,0.8\,L_1 + 0.2\,\text{SSIM}$) [47], which we adopt unchanged: every ablation below isolates the topology channel, never photometric tuning. We add

$$L \;=\; L_{\text{photo}} \;+\; \lambda(t)\, L_{\text{topo}}(V), \qquad (1)$$

where V are the soup's vertex positions and L_{topo} compares the persistence diagram of the live surface to a fixed target bundle $\mathcal{T}$: the significant finite bars of the ground-truth shape's diagram, computed once at the density M the loss will see (alpha values shift with spacing: density matching is a correctness requirement), diagonal-normalized, and locked for all live evaluations. Throughout, a bar is significant iff its lifetime exceeds the measurement floor $6\,r_{\text{med}}(M)$ (r_{med}

= median nearest-neighbour sample spacing; derived and validated in §6).

3.2. Live diagram and pair-frozen backward

At each refresh we draw M surface samples (M=2048 unless the measurement-floor rule of §4 dictates a denser bundle) $X_j = \sum_i w_{ij}\, V[F(f_j), i]$ with faces f_j drawn $\sigma(\alpha)\times$area-weighted and barycentric weights w_{ij} frozen; X is recomputed from live V every step, so gradients reach vertices while the combinatorics stay fixed. The alpha complex of X [12], computed exactly with GUDHI [34], yields persistence pairs (birth simplex, death simplex) per dimension (suppl. Fig. S9). For a Gabriel simplex the alpha filtration value is its squared circumradius, so each paired bar's coordinates are closed-form functions of a few sample positions, all exactly differentiable:

$$R_{\text{edge}} = \tfrac{1}{2}\|p_1 - p_0\|, \qquad R_{\text{tri}} = \frac{\|u\|\,\|v\|\,\|u-v\|}{2\,\|u \times v\|}, \qquad R_{\text{tet}} = \|y\| \quad \text{with} \quad A\,y = \tfrac{1}{2}\,\mathrm{diag}\big(AA^\top\big), \tag{2}$$

where $p_0, \ldots, p_k$ are the k-simplex's sample positions, $u = p_1 - p_0$, $v = p_2 - p_0$, and A stacks the rows $p_i - p_0$ (the circumcenter is $p_0 + y$). Edges carry H0 deaths and H1 births, triangles H1 deaths and H2 births, tetrahedra H2 deaths. We verify, per pairing, that the re-expressed value matches the filtration value (relative 10^{-6}); a pair failing this Gabriel test is detached — it keeps its (correct) value in the loss but propagates no gradient, so an exactness failure can never inject a wrong one. In practice the gate never fires: 100% of significant pairs are Gabriel-exact on all six shapes' clean clouds, and the per-refresh failure count is zero at every one of the 5,500+ logged refreshes (all 2,400 noisy ones included, §5) — a constant pass rate, not an aggregate. Stronger fallbacks (dropping pairs, subgradients, locally raising M) remain available. The pairing is treated as locally constant — the standard pair-frozen device [5, 7] — refreshed every K=10 steps and, necessarily, after every resampling event (which replaces the primitive tensors wholesale); K=10 is an amortization choice, not a tuned knob — it bounds refresh cost (174 ms each; the fixed ≈46 s of §5) while the pairing stays local.

3.3. Matching, diagonal pressure, and recruitment

Let D be the live significant bars and $\mathcal{T}$ the target bars of one dimension. An optimal partial matching (Hungarian on the standard augmented cost; equivalently W_2) splits bars into matched pairs, unmatched live, and unmatched target:

$$L_{\text{topo}} = \sum_{(i,j)\ \text{matched}} \big[(b_i - b_j^*)^2 + (d_i - d_j^*)^2\big] + \sum_{i\ \text{unmatched}} \Big(\tfrac{d_i - b_i}{2}\Big)^2 + L_{\text{recruit}}. \tag{3}$$

The first term pulls matched features to their targets; the second pushes spurious features to the diagonal; the third exists because optimal matching has a provable blind spot:

Lemma 1 If a target bar $t \in \mathcal{T}$ is unmatched under the optimal partial matching, then L_{topo} without L_{recruit} is locally independent of t: no component of its gradient in V acts to create the missing feature.

Proof sketch. The unmatched target's augmented cost is its diagonal penalty $\big((d_t^* - b_t^*)/2\big)^2$, constant in the live coordinates; every V-dependent term of Eq. 3 references live bars only, and the matching is locally constant in V (the pair-frozen regime) — no gradient term is directed at t. □

Recruitment restores a descent direction. Let $\mathcal{T}_{\text{miss}}$ be the unmatched target bars, processed in decreasing persistence order, and let the candidate pool $\mathcal{A}$ hold every live bar not claimed by the matching — unmatched significant bars and, crucially, sub-threshold bars, which matching never sees. Each missing target greedily claims its nearest pool bar, without replacement, under the same squared birth–death metric as the matched term:

$$L_{\text{recruit}} = \sum_{j \in \mathcal{T}_{\text{miss}}} \big[(b_{k(j)} - b_j^*)^2 + (d_{k(j)} - d_j^*)^2\big], \qquad k(j) = \operatorname*{arg\,min}_{k \in \mathcal{A}_j} \big[(b_k - b_j^*)^2 + (d_k - d_j^*)^2\big], \tag{4}$$

where $\mathcal{A}_j$ is the pool remaining after earlier recruitments; recruited bars are excluded from the diagonal term and back-propagate through the same circumradius machinery. This deliberately departs from metric-faithful W_2; §5 probes its location-blindness pathology directly, and C6 shows the term is load-bearing. Greedy rather than optimal assignment is deliberate: in every logged regime the missing-target set is a single bar (torus: one unreached target at every refresh, every seed; §5), where the two coincide; when they diverge, decreasing-persistence order spends the pool on the most significant features first.

3.4. Calibration instead of curriculum

$\lambda(t)$ ramps linearly from 20% to 50% of training, with peak set once by a gradient-norm ratio at the first active step: $\lambda_{\text{peak}} = \rho\, \|\nabla_V L_{\text{photo}}\| / \|\nabla_V L_{\text{topo}}\|$. The single knob ρ is dimensionless (ρ=0.1 throughout); the tail error is flat across $\rho \in \{0.03, 0.1, 0.3\}$, a constant-$\lambda$ ablation matches the ramped version, and alternative ramp windows land within seed noise (suppl. Table S8) — calibration, not scheduling, makes the knob safe.

3.5. Conditions

C0: baseline (photometric only). C1: $+L_{\text{topo}}$. C2: + a short-range repulsion on the same frozen samples, calibrated with the same ρ — the norm-matched control, the RH2 test (C2g: a gentler radius). C3: C1 without the ramp. (C4, curvature-weighted, is deferred; the numbering gap keeps condition tags comparable across studies.) C5: C1 + the best prior of our allocation study (the spread topological field B4; suppl. §A) — the RH3 test. C6: C1 without recruitment (pure matching + diagonal) — does the term matter in-loop, or only for cold-start repair?

4. Experimental setup

Scenes and budgets. Synthetic COLMAP scenes with analytically known topology, rendered from 24–72 Fibonacci-sphere viewpoints at 200–256 px (every-8th test holdout ⇒ 21–63 training views): sphere and cube (void class H2, β=$(1, 0, 1)$, budget N=1200), torus (loop class H1, $(1, 2, 1)$, N=700), two spheres (component class H0, $(2, 0, 2)$, N=700), and genus-2 double torus as a stretch case ($(1, 4, 1)$, N=2000). Budgets are the allocation study's headroom points (suppl. §A) — tight enough that baseline topology is imperfect. Eight external, genus-known meshes probe generality beyond the analytic family, organized as the study's two observable classes (§5.2): a loop group — bob (genus 1), rocker-arm (1), the CGAL eight (2) — and a void group — spot, fandisk, armadillo, horse (all genus 0), the tom-yum pot (3, below); sources, licenses and certificates: suppl. §G. Seven pass the mesh certificate below as-is; five further candidates failed it (open boundaries, non-manifold junctions) and were excluded, never repaired. Membership, observable class and density were fixed by the pre-registered staircase rule before any training (rocker-arm: torus precedent, H1-restricted at M=2048; eight: the pot's, full $(1, 4, 1)$ bundle at M=8192). Scenes: the same pipeline (48 views, 200 px), unchanged class budgets (trio pilot: tails .043–.054, in the analytic band). The fourth, the tom-yum pot (Fig. 1, bottom row; aluminium showcase: suppl. Fig. S7), probes the step every real deployment needs: a raw artist-authored mesh rather than a clean benchmark. As exported from Blender it is textbook triangle soup (9 overlapping open shells, 232 non-manifold junction edges) with no trustworthy ground-truth topology — the same defect that excluded ShapeNet. An ingest-and-certify pass (exact weld, offset-solidification into a thin closed shell, then certification by measurement: exact simplicial homology cross-checked against per-component Euler characteristics, both independent of the loss's sampled alpha complex; watertightness; bit-identical rebuilds) yields one body of genus 3, β=$(1, 6, 1)$ exact, rendered and budgeted as a void-class shape (N=1200). This ingest pipeline is a test fixture here, not a claimed contribution. Where the other externals arrive clean enough to certify as-is, the pot's topology exists only after ingest — a controlled real-world proxy, not a blind test: what real scans will need (mechanics and challenge inventory: suppl. §B, Table S4). The pot also exercises the floor rule of §6 prospectively: every H1 loop of the certified shell sits below the $6\,r_{\text{med}}$ floor at every working M, while the narrow flue mouth caps at small α — an H2 void on the pot axis, first clearing the floor at M=8192 (margin 1.20–1.23×, five sampling seeds). The pre-registered rule builds the bundle there (a floor-rule density deviation later repeated by eight; the loss samples at the same density): exactly one significant bar, stable to re-sampling ($\sim 10^{-5}$).

Protocol. All conditions share one implementation: C0 is the unmodified DiffSoup trainer; the loss enters as one additive term, and disabled it executes none of the added code (no renderer or resampler edits). 2,500 steps; resampling every 100 steps (untouched); ρ=0.1, ramp 0.2:0.5, refresh K=10, M=2048; decisive shapes (sphere, torus) and the double torus run 5 seeds (the former with the full condition set incl. C6), replication shapes (cube, two spheres) 3 seeds (C0/C1/C2). (C7/C7h = C1 + noise; decisive shapes, 3 seeds) perturbs the cloud the loss plan sees with i.i.d. Gaussian noise at every refresh: the plan decides on the noisy observation, the pulls apply to the true live positions. The torus restricts the loss to H1 (its own void bar is subthreshold at M=2048 — a loss must not act on features it cannot reliably measure); the double torus likewise targets its two measurable loops of four. The generality wave runs C0/C1/C2, three seeds, each shape at its staircase-fixed bundle (bob: full loops+void; rocker-arm: H1 only; eight and the pot: M=8192).

Metrics and verdict. Primary: bottleneck to the target diagram in the discriminating dimension, tail-averaged (last three dumps, steps 2,300–2,500), seed-averaged (20k eval samples, target-normalized). Secondary: significant-feature counts (phantom check) and Chamfer parity. Reporting policy: mean±sd over seeds; Welch σ only where both arms ran five seeds; three-seed verdicts rest on disjoint per-seed ranges (suppl. Table S7). Verdict rule: C1 must beat C0 and every norm-matched control at Chamfer parity — a topological win a dumb regularizer can match is not one. Chamfer parity, formally: the passing arm's seed-mean Chamfer must satisfy $\mathrm{Ch}(\mathrm{C1}) \leq 1.15\,\mathrm{Ch}(\mathrm{C0})$; the 15% headroom absorbs seed variance and nondeterminism (§7) and never did any work — every passing arm's Chamfer was at or below baseline.

5. Results

A gate first: the loss alone (no photometric term, Adam on points) repairs all four controlled probes — a punctured void closes (8.8×), a spurious handle dies at the diagonal, mis-spaced components separate (1229×), a bridged merge with no significant live H0 bar (Lemma 1's case) is severed (450×, $\beta_0\,1{\to}2$); the predicted location-blindness did not materialize (suppl. §C, Fig. S5). Raw-cloud sanity toys; multipliers not comparable to training.

5.1. The C-matrix

Table 1 reports tail bottleneck (mean±sd over seeds) in each shape's discriminating dimension; Chamfer in the text where it matters; the full training trajectories for the two decisive shapes are in suppl. Fig. S6 (line = seed mean, band = seed min–max).

Voids (H2): topology-specific, large. Sphere 4.0× (16.2σ Welch vs C0, five seeds), cube 7.9× (three seeds: ranges disjoint, C1 .0064–.0086 vs C0 .0551–.0636), both at equal-or-better Chamfer (sphere 0.46 vs 0.50; cube 0.92 vs 1.05). The norm-matched control is destructive, not merely weaker: it pins the void's death at its repulsion equilibrium (2.2× worse, seed-identical) and erases the cube's void outright; no tested strength reproduces any gain — C1 < C0 < C2g < C2 (§7; spread: suppl. Fig. S3; mechanism: §6).

Loops (H1): the channel's distinguishing result. No prior shape won this class (suppl. §A). The loss cuts torus H1 error 2.3× with zero phantom handles in all five seeds (count trajectories flat at the true $\beta_1{=}2$, suppl. Fig. S4), while the control collapses one loop by step ∼800 and never recovers it. Metric pressure on birth/death values suits 1-D features in a way budget allocation does not (mechanism: §6).

Components (H0): not topological — third channel, same answer. C1 reduces the (already small) baseline error 9.5×, but the generic control matches it (.0008 vs .0007); the increment appears only in Chamfer (0.54 vs 1.94). Three channels agree: β_0 is carried by shell count and separation.

Composition (C5). Loss + spread prior is best everywhere tested — sphere .0082 (7.6× vs C0, ≈2× vs C1 alone, ≈4.4× below the prior alone), torus .0133 — with the best sphere Chamfer (torus .502, trailing only C3's .495). Allocation and value-tuning are complementary, not substitutes.

Stretch case (genus 2): saturated, honestly. Every double-torus arm returns exactly .0264: the two tube loops are unexpressable at feasible budgets — the representation's ceiling, not the channel's (suppl. §B; floors: §6).

Ablations. The ρ-response is flat over a 10× range (suppl. Table S8); C3 (no curriculum) ≈ C1 on both decisive shapes and off the analytic family (bob: C3 .0171±.0026 vs C1 .0208±.0008). Overhead, honestly: the loss adds a fixed ≈46 s per 2,500-step run (refresh 174 ms at $M{=}2048$ every 10 steps) — more than the 33–41 s photometric baseline on these deliberately tiny scenes, but the cost scales $O(M \log M)$ per refresh, not with scene or image complexity, so its share of a run whose photometric cost is T is $46/(T{+}46)$ — an arithmetic projection, ≈7% of a ten-minute run — independent of image count and resolution; no parameters are added.

Recruitment is load-bearing for loops (C6). Removing the recruitment term (pure matching + diagonal) collapses the torus win entirely: .0411 ± .0038, statistically baseline (0.6σ from C0, 12.4σ worse than C1) at baseline Chamfer, though both loops still exist (#sig H1 = 2 every seed; suppl. Fig. S6). The logs locate the mechanism: one torus loop is significant at every refresh, the other chronically below threshold (all five seeds, all 200 refreshes) — recruitment is the second loop's sole gradient path; on this class it is the mechanism, not a cold-start device. On the sphere it never fires (0 of 200, all seeds): C6 is loss-identical to C1, and the 1.5σ tail gap (.0166 vs .0156) measures pure run-to-run noise (§7).

Sensor-noise stress (C7/C7h). With the plan's cloud perturbed at every refresh (σ = 0.5%/1% of the diagonal ≈0.5/1.0 spacings, §4), the loss degrades gradually and never harms: torus .0210±.0018 / .0259±.0022 (2.0×/1.6× below baseline, vs C1's 2.3×), sphere .0235±.0009 / .0328±.0013 (2.7×/1.9×, vs 4.0×), at equal-or-better Chamfer, correct counts in every run (suppl. Fig. S6), zero Gabriel failures in all 2,400 noisy refreshes (internals: suppl. §B).

Table 1. Tail bottleneck to target (mean±sd; lower is better); × = vs C0; verdict: C1 < C0 and < every control at Chamfer parity. Seeds: 5 (sphere, torus, double torus — saturated identically), 3 (cube, two spheres). C6 = no recruitment.

shape (dim)	C0	C1 loss	C2 control	C3 no-ramp	C5 loss+prior	C6 no-recruit
sphere (H2)	.0623±.0063	.0156±.0013 (4.0×)	.1372 (worse)	.0151±.0004	.0082±.0007 (7.6×)	.0166±.0005
cube (H2)	.0582±.0046	.0074±.0011 (7.9×)	.1346 (β_2 1→0)	—	—	—
torus (H1)	.0424±.0031	.0180±.0016 (2.3×)	.0457 (β_1 2→1)	.0155±.0015	.0133±.0004 (3.2×)	.0411±.0038
two spheres (H0)	.0065±.0008	.0007±.0002 (9.5×)	.0008 (ties C1)	—	—	—
double torus (H1)	.0264±.0000	.0264±.0000 (saturated)	.0264±.0000	—	—	—

Table 2. Generality wave, two observable-class groups (mean±sd, 3 seeds; × = vs C0; group mean ± sd over member shapes, pre-registered). Seven of eight pass Table 1's verdict rule; rocker-arm is a no-headroom null (counts intact in every arm). *seed-exact unmatched-target pins (§6). Per-seed values: Table S7; provenance, certificates, staircases: suppl. §G.

shape	C0	C1 loss	C2 control
bob	.0426±.0012	.0208±.0008 (2.1×)	.0437 (β_1 2→1)
rocker-arm	.0083±.0004	.0087±.0015 (0.96×)	.0113 (counts intact)
eight (genus 2)	.0249±.0000*	.0162±.0017 (1.5×)	.0249 (2 of 4 loops)
loop group mean (H1, 3)		1.52× ± 0.55	
spot	.0521±.0081	.0237±.0000 (2.2×)	.0652 (β_2 1→0)
fandisk	.0538±.0022	.0052±.0008 (10.4×)	.0571 (β_2 1→0)
tom-yum pot	.0221±.0003	.0057±.0002 (3.9×)	.0223 (β_2 1→0)
armadillo	.0511±.0000*	.0124±.0001 (4.1×)	.0511 (β_2 1→0)
horse	.0408±.0000*	.0109±.0022 (3.8×)	.0408 (β_2 1→0)
void group mean (H2, 5)		4.87× ± 3.17	

5.2. Generality: external genus-known meshes

Everything above is on one analytic family, so we repeat the verdict protocol — C0/C1/C2, three seeds, class budgets, ρ and ramp unchanged, density fixed by the floor rule — on the eight external genus-known meshes of §4's two groups, with no per-shape tuning. Seven of eight pass the verdict rule at equal-or-better Chamfer (Table 2; original-wave trajectories in suppl. Fig. S2); the eighth, rocker-arm, is a no-headroom null, not a counterexample (obs. 5). The control replicates its failure modes: all five voids erased (β_2 1→0, every seed; the pot at 1.36× worse Chamfer), a bob loop collapsed (β_1 2→1), eight's missing pair left missing.

Observations. (1) The H1 claim survives the family change: bob halves loop error, zero phantom handles every seed (suppl. Fig. S8), replicating torus. (2) The genus-2 ceiling was representational, not topological: on the fat CGAL eight, baseline and control read only two of the four certified loops (value pinned at the unmatched bound, seed-exact); the loss restores the missing pair (#sig 4/3/4) and cuts value error 1.5× at 0.89× Chamfer — the double torus's saturation (§5.1) was the representation's ceiling, not genus-2's. (3) Sharp-featured CAD benefits most: fandisk's 10.4× exceeds every analytic-family effect. (4) Baselines can pin like controls: armadillo and horse read the correct void count but pin its value at the same bound (.0511/.0408, seed-exact); the loss unpins both (4.1×/3.8×) at better Chamfer. (5) No headroom, no effect — and no harm: rocker-arm's one measurable loop is already baseline-correct (.0083 vs the .04–.05 band): §4's imperfection criterion is unmet and the loss stays inert (0.96×, parity 1.00, counts intact everywhere) — the method's honest boundary. (6) Count-level limits persist on sub-budget organic detail: spot's baseline makes a phantom void in two of three seeds (#sig = 2, 1, 2); the loss clears two, retains one (1, 2, 1) — value repair reliable, count repair not guaranteed; its C1 tail pins at .0237 (sd 0), a milder floor of the double-torus kind. And the pipeline reaches raw artist content: the tom-yum pot (Fig. 1; suppl. Fig. S7), a raw Blender export certified rather than constructed and bundled at M=8192 by the same pre-registered rule (§4), cuts the void's value error 3.9× at Chamfer parity (0.99×), the study's tightest seed spread (C1 .0056–.0059 vs C0 .0218–.0223); the control destroys the void (β_2 1→0, all seeds), pinned at the unmatched bound (.0223) — count, not value, is the verdict. Two caveats transfer: its baseline already reads the correct count (a value-accuracy win, like bob), and ground truth is certified, not constructed. Group statistic (pre-registered): loops 1.52 × ±0.55 (2/3 pass), voids 4.87 × ±3.17 (5/5); the passing wave spans 1.5–10.4× (suppl. §G).

6. Mechanism and discussion

A local signal cannot carry a global property — sparse tugs can. A pixel's photometric gradient reaches a few vertices; topology is global and categorical — a wrong genus is not approached smoothly; the blindness probes quantify the gap (§2) — no geometric proxy stands in. The persistence term supplies the measurement with a usable gradient: each significant pair back-propagates through the 3–7 sample positions of its birth and death simplices — precise but sparse pulls, iterated by re-pairing; the resampling prior is the opposite — broad, indirect, budget-limited. The results sort along this axis: loops respond to value pressure (H1 win), wide shells to both channels (composition best), components to neither (H0 null) — and nothing responds where nothing is broken (rocker-arm's no-headroom null, §5.2).

Why the control damages topology. The norm-matched repulsion imposes a length-scale floor on the sampled surface: the void's death pins at that equilibrium (seed-identical) and drags while near-diagonal phantom bars accumulate (suppl. Fig. S1) and thin structures collapse (cube β_2, torus β_1); the loss has no length scale — it pulls the death coordinate itself. The same gradient budget through the same channel without topological information is actively harmful: C1's wins cannot be extra-regularization artifacts.

Measurability at the loss's density. A persistence loss can only enforce features that clear the significance floor at its density M: the torus void and two of the double torus's four loops do not, at M=2048. We treat this as a principle, not a nuisance: the bundle is built and audited at the same M, sub-floor features are excluded, and the double torus is evaluated against its two measurable loops — all the soup can express at feasible budgets.

The floor is quantifiable: significance means lifetime $> 6\,r_{\text{med}}(M)$, and r_{med} follows the uniform-sampling law $\propto M^{-1/2}$ (measured .0090→.0028 over M=2048→20000; ratio 3.16 vs $\sqrt{9.77}$=3.13; suppl. Fig. S10). The double torus's tube loops live at lifetime .045–.046 on the clean surface: at M=2048 that is 0.83–0.85× the floor — invisible by a 15% margin, clearing it only from $M \approx 3\times10^3$ — while at the evaluation density (M=20000, floor .0171) they clear it 3.1×; the binding constraint is the representation (the soup never expresses the tubes): the measurement floor recedes as $M^{-1/2}$, the representation floor is set by N and no M lowers it — the double torus sits on the second. The torus quantifies the noise interaction: its second loop clears the clean floor by only 1.33× — a margin live-cloud noise consumes, leaving recruitment as its only gradient path (C6), the loss reaching below its own floor (suppl. §C). Suppl. Table S3 tabulates counts across M; the pot and the group wave ran the rule prospectively (§4; suppl. §G), validated in-loop (Table 2); adaptive density is the natural extension.

Prescription. Correct topology in the loss; allocate wide; combine — no schedule needed once the loss is calibrated against the photometric gradient.

7. Limitations

Synthetic, single-machine. All quantitative claims are on synthetic renders with known topology: an analytic family plus eight external genus-known meshes in two groups (§5.2; one a certified raw artist mesh, the intended scan path), drawn from public repositories, not a curated benchmark suite. All surfaces are closed except a first open-surface probe, pre-registered and passed: two bowls cut from the sphere (openings $0.14R/0.50R$, certified disk topology) still read one significant void (the designed rim's H1 never clears the floor); the loss cuts its error 4.2×/4.1× at better-than-baseline Chamfer (ranges disjoint); the control erases the wide bowl's void in 2/3 seeds (suppl. §F). C7's Gaussian probe bounds noise response, not scan realism (outliers, missing regions, registration error). Real scans remain future work: their boundaries add noise-born H1 bars (unlike a designed rim); bar-filtering is the near-term step.

Density floor. The loss cannot see features below the significance floor at its density (torus void; two of four double-torus loops at M=2048) — audited, but a real ceiling. Count-level repair of sub-budget detail is likewise out of scope (spot, §5.2).

Fixed target diagram. The target bundle presumes known ground-truth topology (a prior scan or template — not blind capture); template-free designs (total-persistence minimization, a target-genus penalty, class-level diagram priors) are untested.

Control strength. The repulsion control is aggressive (it destroys topology at calibrated magnitude); a half-radius variant is still 1.6× worse (sphere .0972 vs .0623): verdicts do not hinge on control tuning.

Nondeterminism. CUDA training is not bit-reproducible (borderline prune/split decisions flip under float drift at resampling); tail CVs run 7–10% for baseline and loss arms alike, a loss-identical re-run pair (sphere C6 ≡ C1) landed 6% apart — pure noise; statements follow §4's reporting policy.

Recruitment guarantees. The zero-gradient pathology it repairs is provable (Lemma 1); its own self-correction premise is empirical (suppl. §C) — pathological geometries could defeat it; a failure-case study is future work.

8. Conclusion

We moved topology from the allocation channel into the objective: a matched persistence loss with a Gabriel-certified backward and a load-bearing recruitment term, every gain beating a norm-matched control at Chamfer parity. The hypotheses hold (RH1 objective root cause; RH2 topology-specific; RH3 channels compose); verdicts replicate across two external groups and degrade gracefully under noise. Prescription: topology in the loss, allocate wide, combine; next: real scans. Reproducibility: code, scenes, bundles, logs and seeds ship on acceptance (nondeterminism: §7).

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